

M085 Stroke: Hemorrhagic - Expanded Capture Preview

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Benchmark Statistics  Benchmark Statistics

MCG Health
Inpatient & Surgical Care
30th Edition

Stroke: Hemorrhagic

ORG: M-85 (ISC)

Print View

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Care Planning - Inpatient Admission and Alternatives

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Clinical Indications for Admission to Inpatient Care

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- Admission is indicated for **1 or more** of the following(1)(2)(3)(4): [Supporting evidence, suggestions, and alternatives](#)
 - Acute hemorrhagic (eg, intracerebral) stroke(1)(3)(6)

Alternatives to Admission

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- Not usually used

Hospitalization

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Optimal Recovery Course

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Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
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1	• ICU^[A] • Social Determinants of Health Assessment • Adult Readmission Risk Assessment • Discharge planning	• Clinical Indications met^[B]	• Bed rest	• IV fluids • IV medications • Possible central line venous access	• Neurologic checks • Swallowing evaluation • Cardiac monitoring^[C] • Possible mechanical ventilation • Possible arterial catheter • Glucose control • Neurology and neurosurgical consultations • DVT prophylaxis^[D]	• Possible blood pressure control^[E] • Possible anticoagulation reversal • Possible sedatives
2	• Stroke unit or ICU • Social Determinants of Health Assessment • Adult Readmission Risk Assessment	• Neurosurgical indication absent • Hemodynamic stability • Anticoagulant effects absent or reversed • Mechanical ventilation absent	• Activity as tolerated	• IV fluids • IV medications • Possible central line venous access	• Neurologic checks • Possible cardiac monitoring • Possible arterial catheter • Glucose control • Rehabilitation evaluation • Possible physical therapy • Possible occupational and speech therapy • DVT prophylaxis	• Possible blood pressure control • Possible sedatives
3	• Stroke unit or ICU • Social Determinants of Health Assessment • Adult Readmission Risk Assessment	• Mental status at baseline or stable • Neurologic deficits absent or stable • Unimpaired swallowing • Dangerous arrhythmia absent	• Up to chair • Possible assisted ambulation	• Oral hydration • Oral medications • Central lines absent • Advance diet as tolerated	• Arterial catheter absent • Neurologic checks • Glucose control • DVT prophylaxis • Possible physical therapy • Possible occupational and speech therapy	• Sedation absent • Possible blood pressure control
4	• Social Determinants	• Mental status at baseline, or	• Ambulatory or	• Oral hydration^[F]	• Glucose control	• Possible blood pressure

	of Health Assessment • Adult Readmission Risk Assessment • Stroke unit or floor to discharge • Complete discharge planning	stable and acceptable for next level of care • Neurologic deficits absent or stable • Hemodynamic stability • Dangerous arrhythmia absent • Adequate dietary intake • Discharge plans and education understood	acceptable for next level of care	• Oral medications or regimen acceptable for next level of care • Oral diet or acceptable for next level of care • Advance diet as tolerated	• Possible physical therapy • Possible occupational and speech therapy • DVT prophylaxis	control
(1)(3)(5)(6)(8)(10)(16)(17)  Supporting evidence, suggestions, and alternatives  Supporting evidence, suggestions, and alternatives  Supporting evidence, suggestions, and alternatives						

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 3 days

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

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Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for [\(1\)\(8\)\(10\)\(16\)](#):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Surgical intervention [\(1\)\(8\)\(21\)\(22\)](#)
 - Hemorrhage with progressive neurologic compromise may prompt placement of external ventricular drain, surgical hematoma evacuation, or decompressive craniotomy or craniectomy.
 - Anticipate catheter lavage and possible intraventricular thrombolysis for select patients with intraventricular hemorrhage and no underlying vascular lesion. [\(23\)](#)
 - Anticipate possible surgical resection for child with hemorrhagic stroke due to arteriovenous malformation or cavernous malformation. [\[9\]\(4\)](#)
 - Children or adults with focal cerebral arteriopathy (eg, moyamoya) may require surgical revascularization. [\(4\)\(24\)](#)
 - Expect moderate to prolonged stay extension.
 - See [Craniotomy for Traumatic Brain Injury or Intracerebral Hemorrhage](#)
 -  Click here to preview Craniotomy for Traumatic Brain Injury or Intracerebral Hemorrhage^{ISC} guideline.
 - Major deficit or clinical deterioration [\(25\)\(26\)](#)
 - Patient presenting with severe manifestations (eg, hemiplegia, aphasia, progression in neurologic deficits, impaired cognitive or swallowing function) may require ongoing inpatient care.
 - Anticipate use of feeding tube, continuing physical therapy, and possible mechanical ventilation and tracheostomy placement. [\(27\)](#)
 - Expect brief to moderate stay extension.
 - Increased intracranial pressure [\(1\)\(4\)\(6\)\(16\)\(21\)](#)
 - Anticipate hyperosmolar agents (eg, mannitol, hypertonic saline), diuretics, hyperventilation, intracranial pressure monitoring, and possible external ventricular drainage.
 - Decompressive surgical craniectomy may be needed for select patients (eg, supratentorial hemorrhage with midline shift or refractory intracranial hypertension, cerebellar hemorrhage with large hematomas, neurologic deterioration, brainstem compression, or ventricular obstruction). [\(8\)\(16\)](#)
 - Expect moderate stay extension.
 - Hydrocephalus [\(16\)\(21\)](#)
 - Hydrocephalus may require clot evacuation, external ventricular drain placement, or shunt.
 - Expect moderate stay extension.

- Seizures(4)(21)
 - Anticipate parenteral and oral antiseizure therapy and continuous EEG monitoring.
 - Expect brief stay extension.
- Venous thromboembolism(28)
 - Anticipate local external measures and possible vena cava filter.
 - See [Venous Thrombosis and Pulmonary Embolism: Common Complications and Conditions](#)
 Click here to preview Venous Thrombosis and Pulmonary Embolism: Common Complications and Conditions^{ISC} for further information.
 - Expect brief to moderate stay extension.
- Severe electrolyte abnormality (eg, hypernatremia, hyponatremia)(29)
 - Anticipate careful fluid management and close monitoring of electrolytes and mental status.
 - Expect brief stay extension.
- Respiratory failure(16)(27)(30)
 - Intubation and mechanical ventilation may be needed for obtundation, coma (eg, [GCS Calculator](#) GCS Calculator less than 8), loss of protective airway reflexes, significant elevation in intracranial pressure, brain herniation, or non-neurologic respiratory complications (eg, severe aspiration pneumonia, acute respiratory distress syndrome).
 - Tracheostomy may be needed for patient with [Altered mental status that is severe or persistent](#), ongoing need for mechanical ventilation, or failure of extubation.(27)
 - Anticipate brief stay extension.
- Hospital-acquired infection (eg, urinary tract infection, pneumonia, ventriculitis)(28)(31)
 - Anticipate parenteral antibiotics and supportive care.
 - For ventriculitis, anticipate change or removal of intraventricular drain.
 - Expect brief to moderate stay extension.
- Poststroke fever(28)
 - Expect laboratory and radiologic evaluation for infectious and noninfectious etiologies.
 - Anticipate treatment with antipyretic medications and treatment of possible underlying cause (eg, withdrawal of possibly contributing drugs, chest physiotherapy for atelectasis).
 - Expect brief stay extension.
- Poststroke delirium(32)
 - Poststroke delirium is associated with lower poststroke function and higher mortality.
 - Expect brief stay extension.
- Clinically active comorbidities (eg, heart failure, [Acute kidney injury \(stage 2\)](#), [Acute renal failure \(stage 3 acute kidney injury\)](#), hypertension, COPD)(28)
 - Unstable comorbidities may require ongoing inpatient care.
 - Expect brief to moderate stay extension.

See [Common Complications and Conditions](#)  Click here to preview Common Complications and Conditions^{ISC} for further information.

Discharge

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Discharge Planning

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[Expand All / Collapse All]

- Discharge planning includes^[H]:
 - Assessment of needs and planning for care, including(34)(35):
 - Develop and modify treatment plan (involving multiple providers) as needed.
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address psychosocial status issues as indicated. See [Psychosocial Assessment](#)
 Click here to preview Psychosocial Assessment^{SR} for further information.
 - Evaluate and address social determinants of health (eg, housing, food). See [Social Determinants of Health Screening Tool](#)
 Click here to preview Social Determinants of Health Screening Tool^{SR} for further information.
 - Evaluate and address patient or caregiver preferences as indicated.
 - Identify skilled services needed at next level of care, with specific attention to:
 - Home safety assessment(36)
 - Medication management, adherence instruction, and side effects assessment(37)(38)(39)
 - Nutrition and hydration management(40)
 - Ongoing education required(41)
 - Rehabilitation therapy or equipment coordination(42)(43)
 - Early identification of anticipated discharge destination; options include(35)(44):
 - Home; considerations include:

- Home safety assessment. See [Home Safety Assessment](#)  Click here to preview Home Safety Assessment^{SR} for further information.
-  [Collapse](#) Patient safe to go home; examples include(45)(46)(47):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
- Access to follow-up care
- Self-management ability if appropriate. See [Activities of Daily Living \(ADL\) and Instrumental Activities of Daily Living \(IADL\) Assessment](#)  Click here to preview Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment^{SR} for further information.
- Caregiver need, ability, and availability
- Post-acute skilled care or custodial care as indicated. See [Discharge Planning Tool](#)  Click here to preview Discharge Planning Tool^{SR} for further information.
- Transitions of care plan complete, including(35)(44)(48):
 - Patient and caregiver education complete.
 - See [Teach-Back Tool](#)  Click here to preview Teach-Back Tool^{SR} for further information.
 - See [Stroke: Hemorrhagic: Patient Education for Clinicians](#)  Click here to preview Stroke: Hemorrhagic: Patient Education for Clinicians^{SR} for further information.
-  [Collapse](#) Medication reconciliation completion includes(39)(49):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(50)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See [Medication Reconciliation Tool](#)  Click here to preview Medication Reconciliation Tool^{SR} for further information.
- Plan communicated to patient, caregiver, and all members of care team, including(51):
 - Inpatient care and service providers
 - Primary care provider
 - All post-discharge care and service providers
- Appointments planned or scheduled, which may include:
 - Primary care provider(52)
 - Follow-up evaluation call. See [Post-Acute Care Follow-Up Call Tool](#)  Click here to preview Post-Acute Care Follow-Up Call Tool^{SR} for further information.(41)
 - Neurologist(53)
 - Neurosurgeon(3)
 - Psychiatrist or rehabilitation physician(53)
 - Rehabilitation therapy services(43)
 - Other
- Outpatient testing and procedure plans made, which may include:
 - Nuclear medicine(54)
 - Other
- Referrals made for assistance or support, which may include:
 - Alcohol and other drug abuse or dependence treatment
 - Community services
 - Educational program (eg, chronic condition management, self-management)(52)(55)
 - End-stage disease planning(3)

- Financial, for follow-up care, medication, and transportation
- Home architectural modifications
- Self-help or support groups(56)
- Social services (eg, social programs, advance directives)(53)
- Tobacco use treatment(57)(58)(59)
- Vocational rehabilitation(43)
- Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include(60):
 - Adaptive devices (eg, feeding tools, reachers)
 - Ambulation devices (eg, cane, crutches, walker)
 - Antiembolic or compression stockings
 - Bath and toilet aids
 - Blood pressure cuff
 - Chair lift
 - Lift equipment
 - Pressure injury prevention equipment(53)
 - Wheelchairs and accessories
 - Other

Discharge Destination

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- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See [Home Care Indications for Admission Section](#)  Preview disabled, product not licensed^{HC} in Stroke guideline in Home Care.
 - Recovery facility care. See [Recovery Facility Care Indications for Admission Section](#)  Preview disabled, product not licensed^{RFC} in Stroke guideline in Recovery Facility Care.
 - Inpatient rehabilitation facility. See [Inpatient Rehabilitation Facility \(Acute Rehabilitation\) Indications for Admission Section](#)  Preview disabled, product not licensed^{RFC} in Inpatient Rehabilitation Facility (Acute Rehabilitation): Stroke guideline in Recovery Facility Care.

Evidence Summary

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Criteria

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The evidence for the clinical indications found in this guideline includes 2 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, and 1 book section.

Analysis of an all-payer database shows a mean length of stay of 8.1 days for patients with a principal diagnosis of hemorrhagic stroke admitted to inpatient care.(5) (EG 3) A consensus guideline and a narrative review state that patients with hemorrhagic stroke require admission to inpatient care, often to the neurocritical care or dedicated stroke unit.(1)(3) (EG 2)

Hospitalization

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A systematic review and meta-analysis of 14 randomized controlled trials (1957 patients) found that, compared with conventional care, hospitalized stroke patients given early supported discharge services had lower rates of long-term disability or institutionalization and shorter lengths of stay.(18) (EG 1) A meta-analysis of 8 studies encompassing 2608 patients with acute ischemic or hemorrhagic stroke randomized to very early mobilization (intensive out-of-bed activities within 2 days of onset) or usual care found early mobilization shortened hospital stay (mean weighted difference 1.7 days) and improved activities of daily living, with no increase in mobilization-related adverse events.(19) (EG 1)

Readmission risk and reduction: Analysis of 8279 patients with acute ischemic or hemorrhagic stroke (mean age 70 years) found, after statistical adjustment, that treatment by physical or occupational therapists was independently associated with a lower risk of the composite outcome of readmission or death at both 30-day and 7-day time periods.(20) (EG 2)

Length of Stay

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Analysis of a database of 597,046 adults admitted with nontraumatic intracerebral hemorrhage found that 25% were discharged in 2 days or less.(17) (EG 2) Analysis of national hospital discharge data shows 31% of hospitalized patients with the principal diagnosis of nontraumatic intracerebral hemorrhage were discharged in 3 days or less.(5) (EG 3)

Rationale

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Use of this MCG care guideline helps the clinician identify patient-specific complex clinical factors that make it reasonable to expect a necessity for hospital care across 2 or more midnights. The evidence-based clinical criteria assist the clinician in the decision to appropriately admit a patient to inpatient care. For Medicare enrollees, MCG care guidelines also support the clinician in the decision to appropriately admit a patient to inpatient care when there is not an expectation of 2 or more midnights of hospital care (eg, CMS' Two-Midnight Rule exceptions).

Use of these evidence-based clinical criteria to support decision making around the need for inpatient treatment is of clinical benefit to the patient and is crucial for quality patient care. Use of evidence-based clinical criteria ensures patients receive the most appropriate care for their specific condition, reduces unnecessary risks, promotes recovery, and provides a standard that can reduce disparities in care. Evidence-based clinical criteria not only help align treatment decisions with best practice, but they can also help reduce unwarranted variation in care by fostering consistency and equality in healthcare provision across regions and facilities. Appropriate use of the evidence-based clinical criteria in conjunction with the Supplemental Medicare Criteria ensures Medicare patients receive full access to Medicare benefits (eg, inpatient care), without risk of delay or decreased access, based on variables such as clinical severity of illness, length of provision of medically necessary hospital-level care, or satisfaction of specified Medicare criteria as directed by CMS. Use of these criteria will help ensure that the patient receives necessary care in the appropriate setting.

Related CMS Coverage Guidance

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This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(7); 42 CFR 419.22(61); 42 CFR 422.101(62)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(63); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(64); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(65); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(66)

Medicare Coverage Determinations: Medicare Coverage Database(67)

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Footnotes

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[A] ICU is indicated for all patients with acute spontaneous intracerebral hemorrhage.(1)(8)(9)(10) If available, a neurocritical care unit with neurosurgical capabilities is preferred.(1)(11)(12)(13) See [Intensive, Intermediate, and Telemetry Care Guidelines](#)

 Click here to preview Intensive, Intermediate, and Telemetry Care Guidelines^{ISC}. [A in Context Link 1]

[B] See [Clinical Indications for Admission to Inpatient Care](#) in this guideline. [B in Context Link 1]

[C] Cardiac arrhythmias may occur.(1) [C in Context Link 1]

[D] Intermittent compression devices should be given for DVT prophylaxis on admission. Pharmacologic prophylaxis may be considered after 24 to 48 hours of stability.(1)(14) [D in Context Link 1]

[E] A specialty society guideline recommends careful titration of antihypertensive therapy for patients with mild to moderate spontaneous intracerebral hemorrhage whose SBP is between 150 and 220 mm Hg, with a goal of initiating treatment within 2 hours of onset and reaching a target range of 130 to 150 mm Hg.(1) The same guideline recommends against acutely lowering SBP to less than 130 mm Hg because of increased risks of acute renal injury and other adverse events.(1) This same guideline finds insufficient data to make recommendations for patients with SBP greater than 220 mm Hg or for large intracranial hemorrhage.(1) However, another guideline concludes, based on low-quality data, that the careful use of continuous IV medication under close blood pressure monitoring to lower blood pressure is reasonable for patients with SBP greater than 220 mm Hg.(15) Treatment of hypertension is also recommended if there is stroke with concomitant acute myocardial ischemia, aortic dissection, preeclampsia, or eclampsia.(16) [E in Context Link 1]

[F] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [F in Context Link 1]

[G] For children with hemorrhagic stroke due to arteriovenous malformation, surgical resection or radiation therapy is preferred over embolization therapy due to high rates of hemorrhage post embolization.(4) Embolization is preferred as first-line therapy for most cases of arteriovenous fistula.(4) [G in Context Link 1]

[H] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(33) [H in Context Link 1]

Codes

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ICD-10 Diagnosis: I61.0, I61.1, I61.2, I61.3, I61.4, I61.5, I61.6, I61.8, I61.9

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